Reliable Knowledge-Grounded AI Copilots for Dynamic Information Synthesis

Overview AI copilots based on retrieval-augmented generation can help synthesize information from large collections of reports, but retrieving apparently relevant documents does not guarantee a sound synthesis. A copilot may conflate similarly named entities, combine obsolete and current observations, omit a critical qualification, or infer a conclusion that does not follow from individually accurate facts. These errors are especially difficult to detect when information sources are heterogeneous, partially contradictory, and continuously updated. Standard self-critique often produces another fluent answer without establishing which claims are supported.

We propose grounded AI copilots that build structured knowledge from changing information, decompose generated conclusions into verifiable assertions, and selectively repair unsupported claims. The project targets three methodological challenges: (1) reconciling evidence from heterogeneous sources while retaining temporal and provenance information; (2) determining which verification operations and scientific or logical rules apply to each synthesized claim; and (3) using fine-grained feedback to correct an output without damaging its already supported components.

We will pursue three interrelated research threads:

- **Structured integration of heterogeneous evidence:** We will extract entities, attributes, events, relations, and source-level provenance from long and evolving documents. Building on our research in consistent event-graph learning, we will investigate representations that connect reports across time and explicitly track duplicate, contradictory, and superseded assertions. The copilot will update these representations as new documents arrive rather than repeatedly synthesize from an unstructured collection.
- **Multi-step claim verification with domain knowledge:** We will decompose a generated synthesis into atomic claims and infer the prerequisites needed to justify each claim: evidence passages, graph relations, numerical constraints, or domain principles. A verification planner will select among retrieval, knowledge-graph queries, symbolic checks, and external tools. Following our FAIGEN approach, a rule will be considered applicable only when its conditions are satisfied; the copilot will report supported, contradicted, and unresolved claims separately.
- **Fine-grained feedback and selective self-correction:** We will develop error feedback over entities, attributes, relations, factual grounding, and logical consistency. Synthetic perturbations and independently checked examples will provide supervision for diagnosing unsupported reasoning steps. A revision policy will request missing evidence and modify only the affected portions of the synthesis, followed by checks for unintended downstream inconsistencies and regressions.

Our evaluation will include controlled cases where two individually correct retrieved sentences yield a false combined conclusion because they concern different times, entities, or operating assumptions. Other cases will replace an authoritative document with a newer correction or introduce a legitimate ambiguity that should remain unresolved. We will examine whether the copilot knows when to abstain, which source it chooses to verify, and whether its repairs preserve correct portions of the initial synthesis. Component ablations will identify when structured knowledge and rule-aware checks provide advantages beyond repeated prompting or larger retrieval contexts.

Keywords: AI copilots, Information synthesis, Retrieval augmentation, Fine-grained verification.

Intellectual Merit The project will advance structured dynamic retrieval, rule-aware verification, and localized correction rather than treating synthesis as unconstrained text generation. We will benchmark factual support, consistency, update sensitivity, selective-repair accuracy, and computational cost under controlled stale and conflicting reports. Comparisons include conventional RAG, iterative retrieval, and LLM self-critique.

Naval Relevance and Broader Impact Potential Naval applications include trustworthy assistance for logistics, maintenance, and situation-report synthesis, where users must distinguish current evidence from superseded or unresolved information. Research artifacts will use public or synthetic records, avoiding any assumption of privileged operational datasets.